COMS 4720/5720 Principles of Artificial Intelligence

Midterm Exam

Fall 2024

Name: _____

ISU NetID (username): _____@iastate.edu

1	2	3	4	5	6	Total
30	8	10	16	18	18	100

1. **[30 pts]** *Short Questions*

(a) [**6 pts**] Determine if the following statements are true or false. For each statement, mark only the answer you think is correct.

(i) The number of belief states is exponential in the number of physical states in the worst case.

true _____ false _____

(ii) In Monte Carlo tree search, the search tree for the current state grows by exactly one node for every playout.

true _____ false _____

(iii) The memory requirement for the iterative deepening search is linear in the depth d at which the goal state is found.

true _____ false _____

(iv) A best-first search terminates as soon as a goal state is found.

true _____ false _____

(v) The A* algorithm using an inadmissible heuristic can never find an optimal solution path.

true _____ false _____

(vi) The minimax algorithm performs a breadth-first search in the game tree.

true _____ false _____

(b) [**6 pts**] Fill answers (Yes / No) in the following table which compares uninformed search algorithms. Assumptions: (i) the branching factor is finite; (ii) the state space either has a solution or is finite; and (iii) all action costs are equal to the same positive constant.

	Uniform-cost	Iterative deepening	A* using a zero g-cost function
Complete?			
Optimal cost?			

(c) [**4 pts**] Consider a search employing some heuristic function h. Let n and n' be two nodes in the search tree such that n' is generated from n by applying some action a. Suppose that $h(n) = 12$ and $h(n') = 8$. The cost of a is 3, denoted as $c(n, a, n') = 3$. Is the used heuristic consistent? Explain your answer in one sentence.

(d) [**6 pts**] Both breadth-first search (BFS) and depth-first search (DFS) are conducted in a state space. Let b be the branching factor, d be the depth of the (unique) goal state, and m be the maximum depth of the resulting search tree. In big-O notation give the time and space complexities of the two algorithms.

BFS runs in $O($ $\qquad\qquad$ $)$ time and requires $O($ $\qquad\qquad$ $)$ memory.

DFS runs in $O($ $\qquad\qquad$ $)$ time and requires $O($ $\qquad\qquad$ $)$ memory.

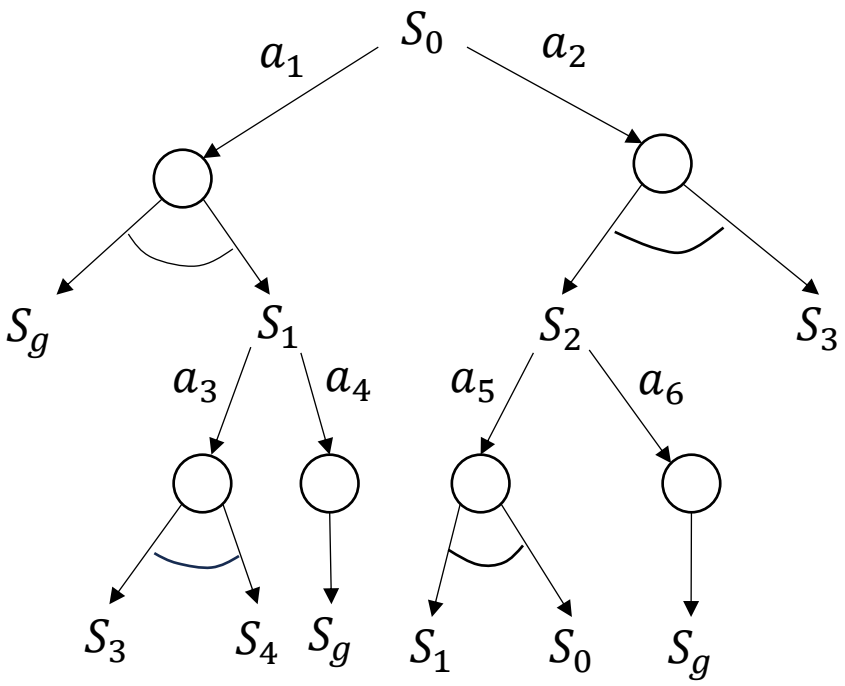
(e) [**4 pts**] List two situations where the Monte Carlo tree search is preferred to the alpha-beta tree search with pruning.

(f) [**4 pts**] List two techniques used by an evolutionary algorithm in producing offsprings to form the next generation.

2. **[8 pts]** *Search with Nondeterministic Actions*

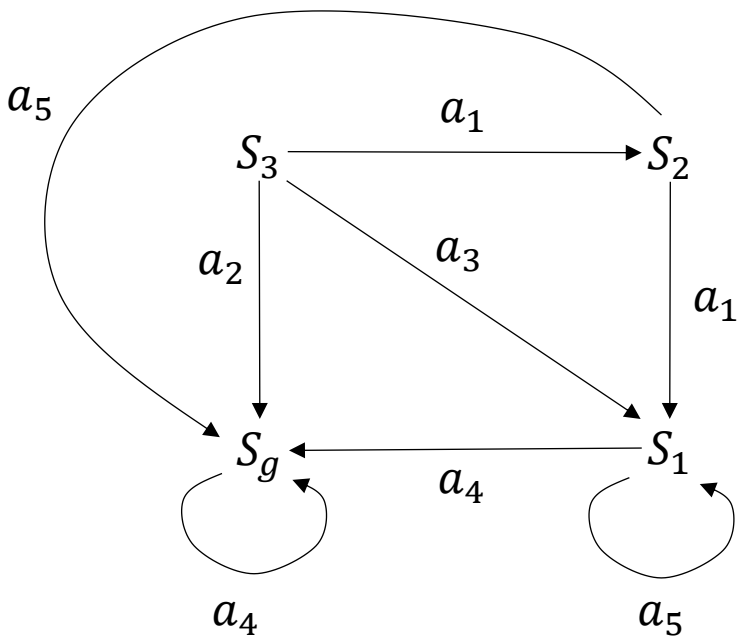
Shown below is an AND-OR search tree, where $S_0, S_1, S_2, S_3, S_4, S_g$ are six states and $a_1, a_2, a_3, a_4, a_5, a_6$ are six actions. Here, S_0 is the initial state and S_g is the goal state. Note that (i) some of the actions may be erratic in non-deterministically generating an outcome state and (ii) every state with no children shown is a goal state, has no applicable actions, or reappears to form a loop.

Draw a closed contour around the subtree that represents a solution. You may also copy the solution subtree over to the right of the search tree.



3. **[10 pts]** *Search with No Sensing*

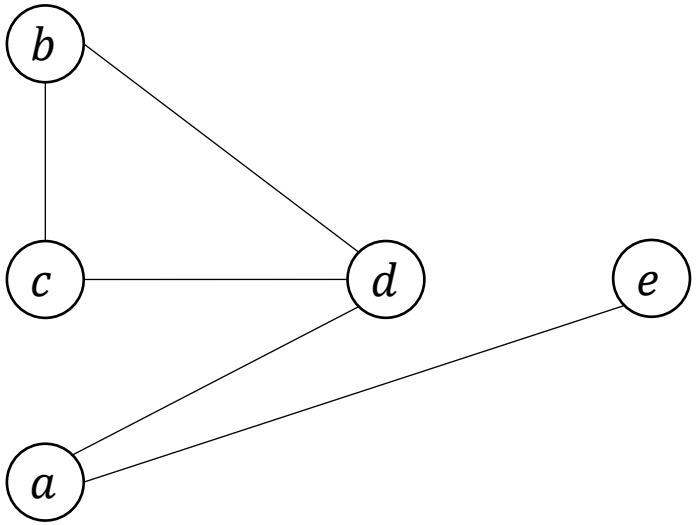
Consider a search problem that has four physical states S_1, S_2, S_3, and S_g, the last of which is the goal state. There are five actions a_1, a_2, a_3, a_4, and a_5. The state transition model is described by a directed graph below. Suppose that (i) no sensor is available and (ii) the agent is aware that at the start the physical state must be either S_2 or S_3.



(a) [**2 pts**] Give the initial belief state.

(b) [**8 pts**] Does there exist a solution path from the initial belief state to the goal belief state $\{S_g\}$? If so, in the below draw the solution path on which nodes are belief states and directed edges are labelled with actions.

4. **[16 pts]** *Constraint Satisfaction*



Given the graph on the left, we would like to color its five nodes in *red* (R), green (G), and *Blue* (B) such that no two adjacent nodes are assigned the same color. At the start, the domains D_a, D_b, D_c, D_d, D_e of the five nodes a, b, c, d, e , respectively, have the same value $\{R, G, B\}$. During the coloring, some domains may be reduced.

(a) [**2 pts**] According to the degree heuristic, which node is to be colored first?

(b) [**10 pts**] Suppose that the color B has been applied to the first node chosen in (a). Write down the (possibly reduced) domain of every variable, as a result of this first assignment, to keep the graph arc-consistent.

$D_a = \{ \qquad \}$ $\qquad$ $D_b = \{ \qquad \}$

$D_c = \{ \qquad \}$ $\qquad$ $D_d = \{ \qquad \}$

$D_e = \{ \qquad \}$

(c) [**4 pts**] Assume the choice of the first node in (a) and its color assignment in (b). How many coloring schemes (i.e., complete and consistent assignments) exist? (No need to write them down.)

5. **[18 pts]** *A* search and the 8-puzzle*

6	8	
4	5	3
7	1	2

Initial State S_0

1	2	3
5	8	4
7		6

Goal state S_g

An 8-puzzle has its initial state S_0 and goal state S_g shown on the left. To solve the puzzle, we are equipped with two heuristic functions: $h_1(S)$, which is the number of misplaced tiles in a state S when compared to the goal state S_g, and $h_2(S)$, which sums up the Manhattan distance between the position of every tile in S and its position in S_g.

(a) [**6 pts**] The number of inversions in the initial state S_0 is _____.

The number of inversions in the goal state S_g is _____.

(b) [**2 pts**] Is this 8-puzzle solvable? Why or why not?

(c) [**8 pts**] Suppose that the agent did not check the solvability of this 8-puzzle but still tries to solve it using the A* algorithm. There are two moves to choose from at the state S_0: *Right* and *Up*, which respectively yield the states S_1 and S_2, as shown below in the search tree. Evaluate the two heuristics h_1 and h_2 first on S_1 and then on S_2.

Initial State S_0

6	8	
4	5	3
7	1	2

Goal state S_g

1	2	3
5	8	4
7		6

Right

6		8
4	5	3
7	1	2

S_1

Up

6	8	3
4	5	
7	1	2

S_2

$h_1(S_1) =$

$h_2(S_1) =$

$h_1(S_2) =$

$h_2(S_2) =$

(d) [**2 pts**] Suppose that the agent decides to use the number h_1 of misplaced tiles as the heuristic. Which of the two moves will be chosen as the first move?

6. **[18 pts]** *Alpha-Beta Pruning*

You are given a minimax search tree shown on the next page. The tree has ten internal nodes $A, B, \ldots, J$. Not all terminal states (leaves) are at the same depth. Execute the alpha-beta pruning algorithm at the bottom of this page. (It is the same algorithm on the PowerPoint slides presented in class.)

(a) [**8 pts**] Mark all the subtrees (including leaves) that have been pruned. You may, for instance, simply put double slashes \\ or // across the edge entering the root of such a subtree from the above.

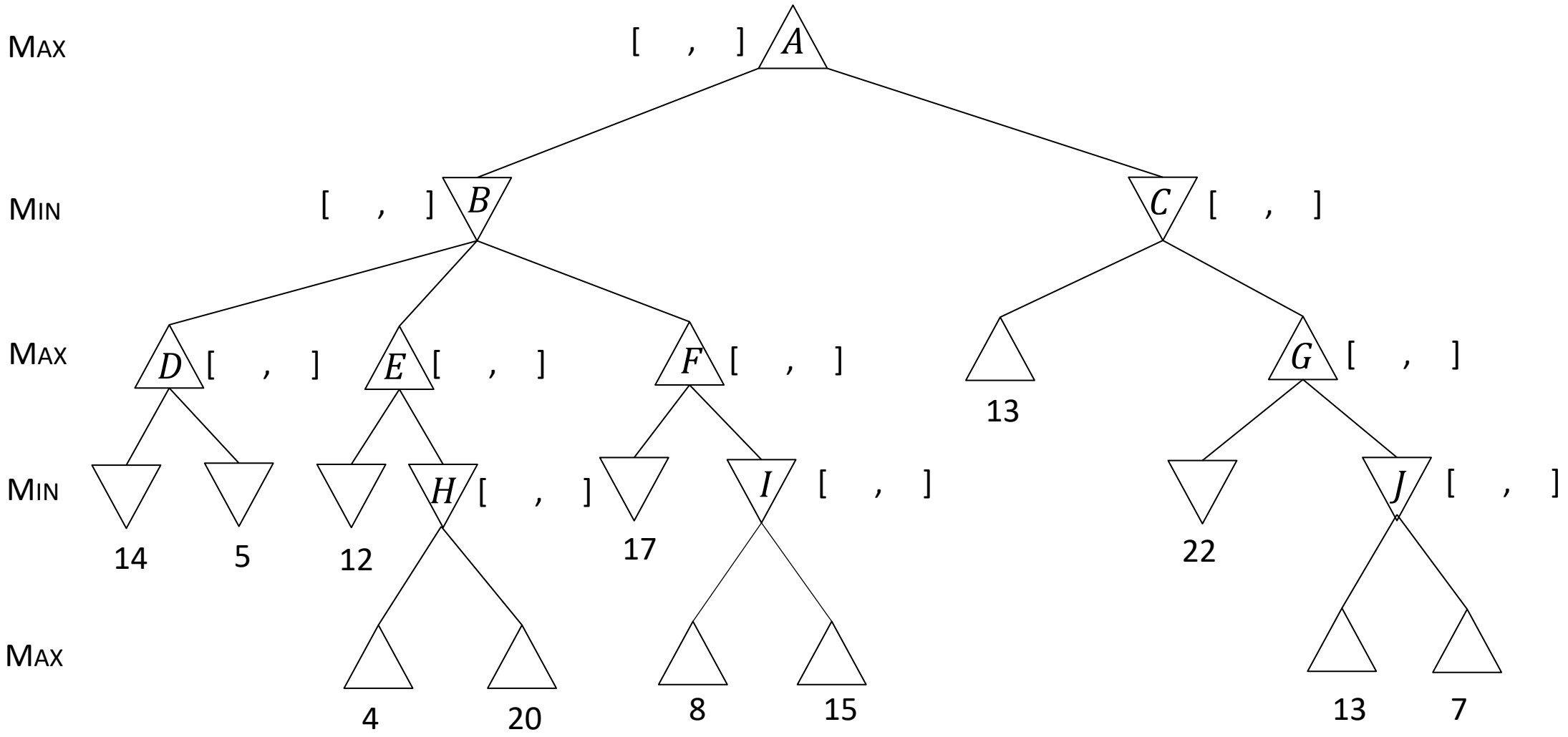
(b) [**8 pts**] At every internal node storing a state on which a call MAX-VALUE or MIN-VALUE is invoked, fill inside the bracket [,] next to the node the values of α and β at the completion of this call.

(c) [**2 pts**] What is the final value for MAX at the root?

```
function ALPHA-BETA-SEARCH(state) returns an action
  v ←MAX-VALUE(state, −∞, +∞)
  return the action in ACTIONS(state) with value v
```

```
function MAX-VALUE (state,α, β) returns a utility value
  if TERMINAL-TEST(state) then return UTILITY(state)
  v ← −∞
  for each a in ACTIONS(state) do
    v ←MAX(v, MIN-VALUE(RESULT(s, a),α, β))
    if v ≥ β then return v
    α ←MAX(α, v)
  return v
```

```
function MIN-VALUE (state,α, β) returns a utility value
  if TERMINAL-TEST(state) then return UTILITY(state)
  v ← +∞
  for each a in ACTIONS(state) do
    v ←MIN(v, MAX-VALUE(RESULT(s, a),α, β))
    if v ≤ α then return v
    β ←MIN(β, v)
  return v
```


MAX
[,]
A
MIN
[,]
B
C
[,]
MAX
D
[,]
E
[,]
F
[,]
13
G
[,]
MIN
14
5
12
H
[,]
17
I
[,]
22
J
[,]
MAX
4
20
8
15
13
7

(Scratch only)

(Scratch only)

(Scratch only)